

EXECUTIVE BRIEFING

AI Agents & Agentic AI

From models that answer to systems that act

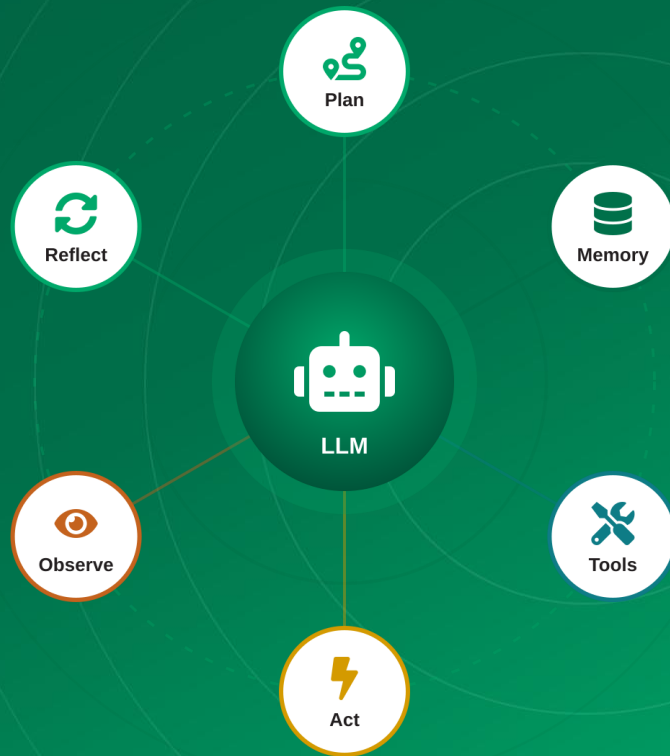
Concepts

Architectures

Ecosystem

Risks

Data Innovation For Africa (DI4A) - African Development Bank



What we will cover

01



The shift

Why AI is moving from answering questions to completing tasks

02



Definitions

Agent vs. agentic, and the autonomy spectrum

03



How agents work

Anatomy, the agent loop, reasoning and memory

04



Architectures

Tools & protocols, workflow patterns, multi-agent systems

05



State of the field

Milestones, ecosystem, benchmarks and momentum

06



Doing it right

Risks, best practices and a decision guide

SECTION

01

The shift

Why is AI moving from answering to acting?



From answering to acting



Generative AI assistant

e.g. a classic chatbot

TRIGGER	Responds to a single prompt
PROCESS	One pass: prompt → answer
OUTPUT	Text, code or images to review
WHO ACTS?	The human does the work



Agentic AI system

e.g. a coding or research agent

TRIGGER	Pursues a goal
PROCESS	Multi-step loop: plan → act → observe
OUTPUT	Completed task & changed state
WHO ACTS?	The agent, under human oversight

Generative AI creates content. Agentic AI gets tasks done.

SECTION

02

Definitions

What exactly is an agent — and what is not?



What is an AI agent?

CLASSICAL AI

“Anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.”

— Russell & Norvig, Artificial Intelligence: A Modern Approach

LLM ERA

Systems where the LLM dynamically directs its own process and tool usage, keeping control over how it accomplishes the task.

— Anthropic, “Building Effective Agents” (Dec 2024)

In practice: an LLM + tools + memory, running in a loop toward a goal.

KEY PROPERTIES



Autonomy

Decides the next step without a script



Goal-directed

Works toward an outcome, not a single reply



Tool use

Calls APIs, runs code, searches, clicks



Memory

Keeps context within and across tasks



Adaptation

Observes results and corrects course

Agent vs. agentic: a matter of degree

AGENT (*noun*)

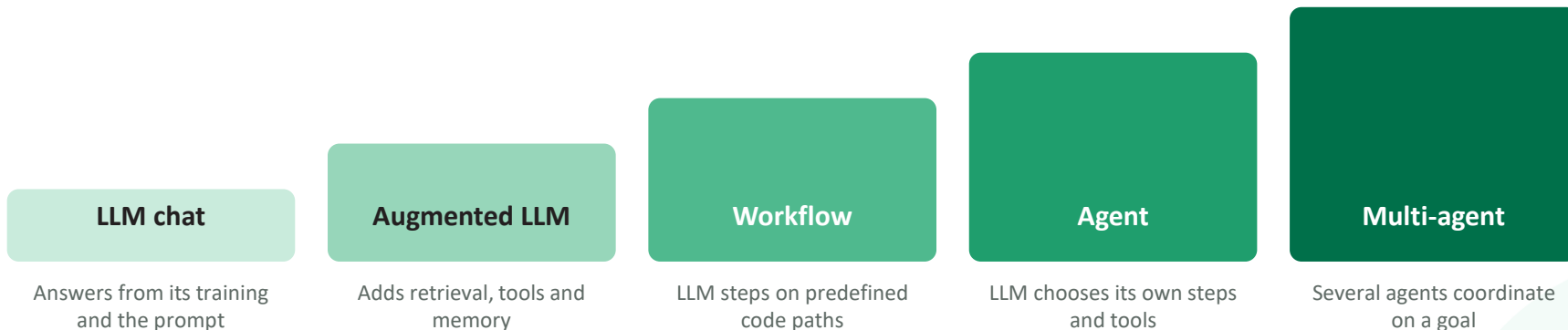
A system that perceives, decides and acts to reach a goal — the “what”.

AGENTIC (*adjective*)

The degree of autonomy a system has. Popularised by Andrew Ng (2024) to move past the yes/no debate.

THE AUTONOMY SPECTRUM

More autonomy = more flexibility... and more cost, latency & risk



SECTION

03

How agents work

What happens inside an agent?



Not everything that automates is an agent



RPA bot



AI assistant



AI agent

HOW IT WORKS

Follows fixed rules and scripts

Answers prompts in natural language

Reasons, plans and uses tools in a loop

UNEXPECTED INPUT

Breaks when screens or formats change

Understands it, but only replies

Adapts its plan to what it observes

WHO DECIDES THE STEPS

The developer, in advance

The human, turn by turn

The agent, within guardrails

BEST FOR

Stable, high-volume, rule-based tasks

Q&A, drafting, summarising

Variable multi-step tasks with clear goals

EXAMPLE

Copying invoice data into an ERP

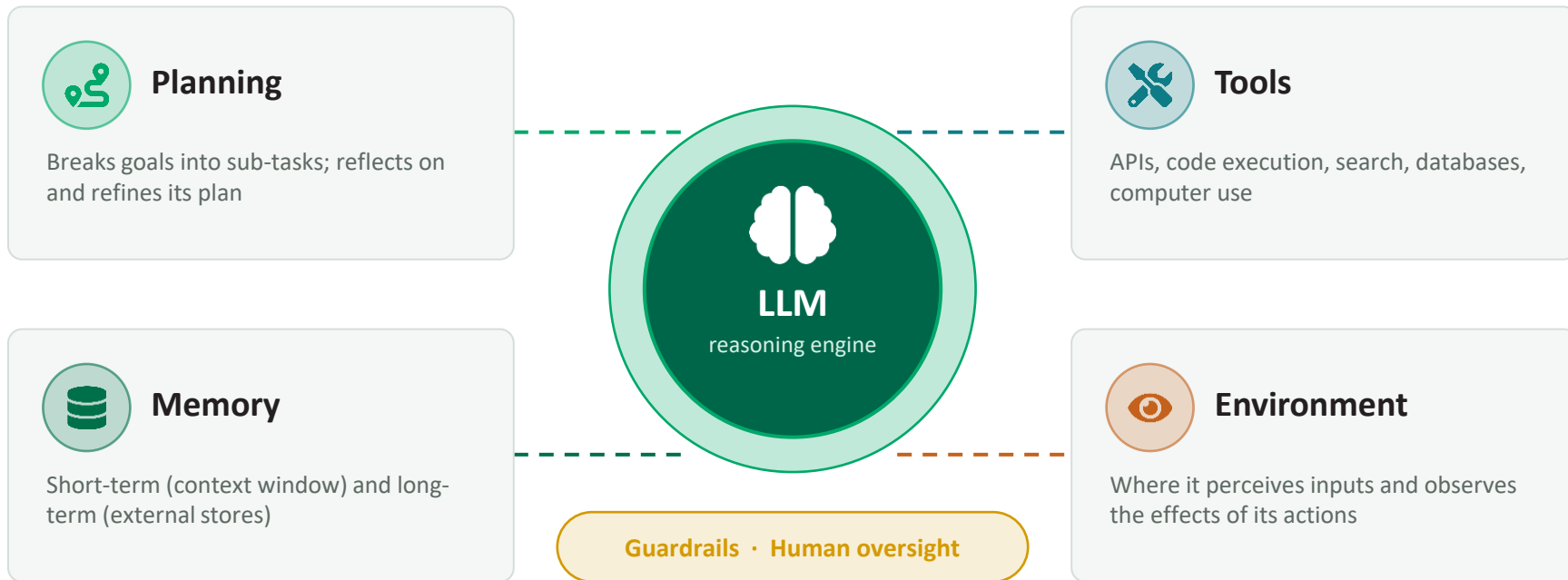
Explaining an HR policy

Resolving a support ticket end-to-end



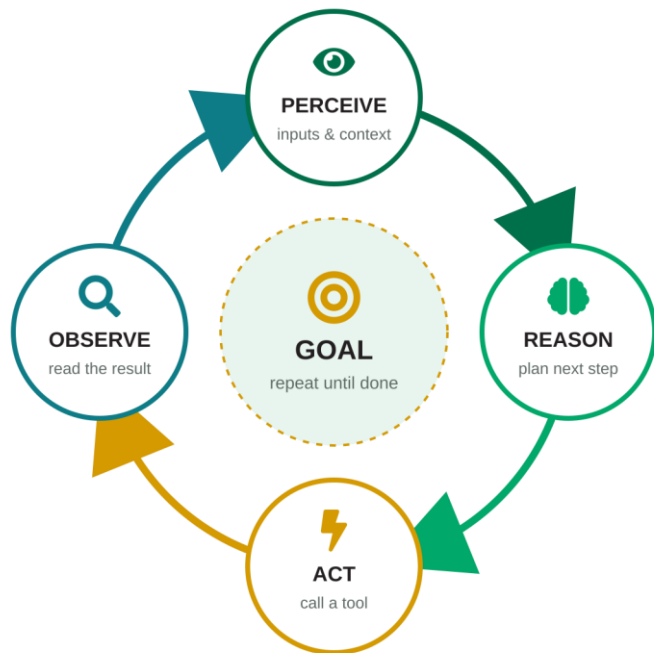
Beware of “agent washing”: rebranding assistants, chatbots or RPA as agents without real agentic capabilities (Gartner).

Anatomy of an AI agent



Component view adapted from Lilian Weng, "LLM Powered Autonomous Agents" (2023)

The agent loop



REACT IN ACTION · illustrative trace

Goal: “Remind Acme about their overdue invoice”

Thought Find the latest unpaid invoice first.

Action `search_invoices(client="Acme")`

Observe INV-2291 · €12,400 · 34 days overdue

Thought Draft a polite reminder with these details.

Action `create_email_draft(to="billing@acme...")`

Observe Draft saved → request human approval

The loop stops when the goal is met, a limit is reached (steps, time, budget), or human input is needed.

Reasoning & planning techniques

Chain-of-Thought

2022

Reason step by step before answering; improves multi-step problem solving.

Wei et al.

ReAct

2022

Interleave reasoning with actions and observations — the basis of most agents.

Yao et al.

Reflexion

2023

Self-critique after feedback; lessons stored in memory for the next attempt.

Shinn et al.

Tree of Thoughts

2023

Explore several reasoning paths, evaluate them and backtrack.

Yao et al.

Plan-and-Solve

2023

First devise a plan of sub-tasks, then execute them one by one.

Wang et al.

Reasoning models

2024

Models trained to think at length before acting (e.g. OpenAI o1, Sept 2024).

Industry

Memory: how agents remember

TYPE	WHAT IT HOLDS	TYPICAL IMPLEMENTATION	HUMAN ANALOGY
 Working	The current task: instructions, recent steps, tool results	The context window	<i>What you hold in mind right now</i>
 Episodic	Past interactions and experiences	Logs, conversation history, reflections	<i>Remembering last week's meeting</i>
 Semantic	Facts and knowledge about the world or the business	Vector databases, RAG, knowledge graphs	<i>Knowing that Paris is in France</i>
 Procedural	Skills and how to do things	Model weights, prompts, code, reusable skills	<i>Knowing how to ride a bike</i>

Taxonomy from CoALA — Sumers et al., “Cognitive Architectures for Language Agents” (2023)

SECTION

04

Architectures

How are agentic systems built?



Tools & protocols: connecting agents to the world



Function calling

The model emits a structured tool call; the application runs it and returns the result.



Model Context Protocol (MCP)

Open standard to connect agents to tools and data. Anthropic, Nov 2024; donated to the Linux Foundation's Agentic AI Foundation in Dec 2025.



Agent2Agent (A2A)

Open protocol for agents to discover and collaborate with other agents. Google, Apr 2025; now a Linux Foundation project.

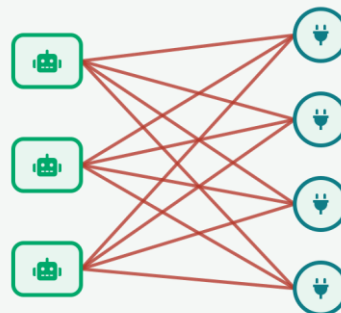


Computer use

Agents operate graphical interfaces via screenshots, mouse and keyboard (Anthropic, Oct 2024; OpenAI Operator, Jan 2025, now part of ChatGPT agent).

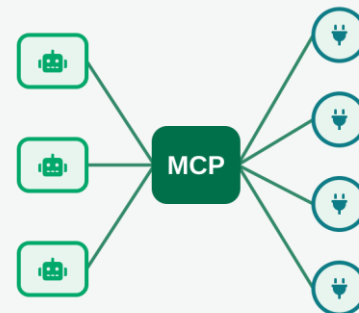
WHY A STANDARD MATTERS

Without a standard



$M \times N$ custom integrations

With MCP



$M + N$ — build once, reuse

MCP = agent ↔ tools & data

A2A = agent ↔ agent

Building blocks: workflows vs. agents



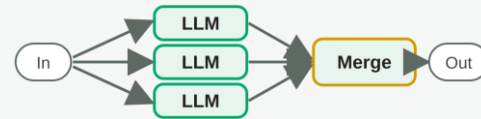
Prompt chaining

Fixed sequence of LLM calls, with optional checks (gates) between steps.



Routing

Classify the input, then send it to the most suitable specialised handler.



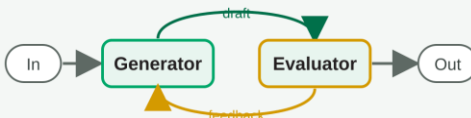
Parallelization

Run sub-tasks simultaneously (sectioning) or several times (voting), then merge.



Orchestrator-workers

A central LLM breaks down the task dynamically and delegates to workers.



Evaluator-optimizer

One LLM generates, another critiques, in a loop until quality is met.

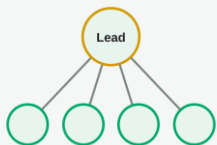


Autonomous agent

The LLM decides its own steps, using environment feedback, until it stops.

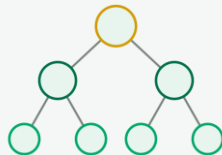
WORKFLOWS follow predefined code paths — predictable. **AGENTS** direct themselves — flexible. Source: Anthropic (2024)

Multi-agent systems



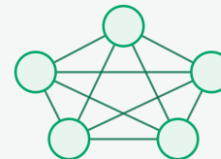
Orchestrator-subagents

A lead agent plans and delegates to specialists working in parallel



Hierarchical

Managers of managers — teams of agents organised in layers



Peer network

Agents hand off or negotiate directly with one another

+90.2%

Anthropic's multi-agent research system (Claude Opus 4 lead + Sonnet 4 subagents) vs. a single Opus 4 agent on an internal research eval (2025)

Relative token usage (chat = 1x)



SECTION

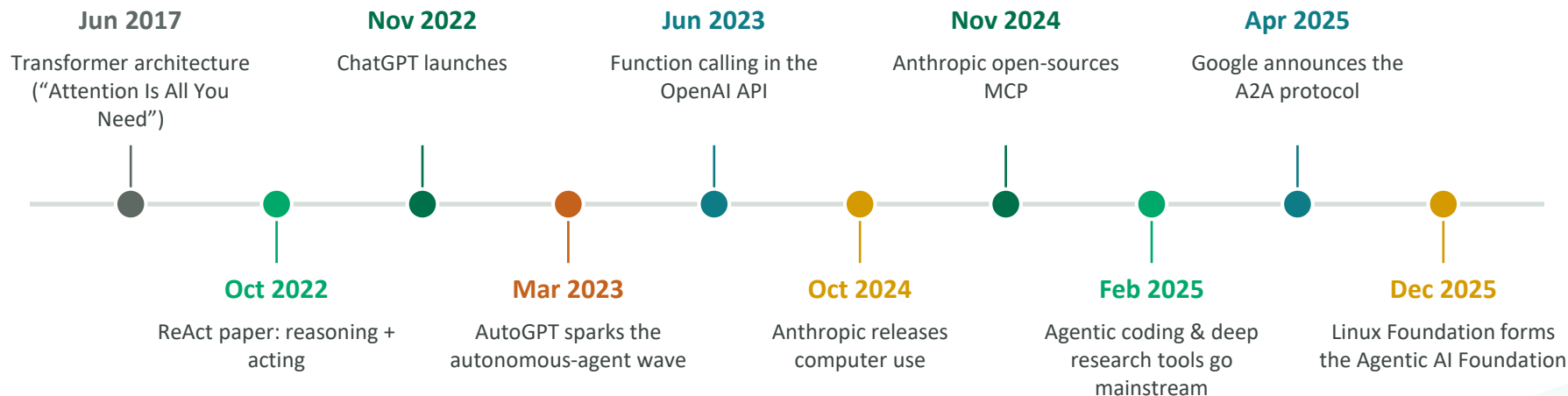
05

State of the field

Where do we stand today?



Key milestones of the agentic era



Feb 2025 examples: Anthropic Claude Code (research preview) and OpenAI Deep Research.

The ecosystem at a glance



Open-source frameworks

- LangGraph (LangChain)
- CrewAI
- Microsoft Agent Framework (successor to AutoGen & Semantic Kernel)
- LlamaIndex
- goose (now under AAIF)



Vendor agent SDKs

- OpenAI Agents SDK
- Claude Agent SDK (Anthropic)
- Agent Development Kit (Google)
- Amazon Bedrock Agents



Agent products

- Coding: Claude Code, OpenAI Codex, GitHub Copilot, Cursor
- Research: Deep Research (OpenAI, Google)
- Enterprise: Salesforce Agentforce, Microsoft Copilot Studio



Open standards

- MCP — agent ↔ tools
- A2A — agent ↔ agent
- AGENTS.md — guidance for coding agents

Non-exhaustive selection — the landscape evolves rapidly.

Measuring agents: benchmarks & progress

SWE-bench Verified

Fix real GitHub issues (500 human-validated tasks)

SWE-bench Pro

Harder, more contamination-resistant successor (Scale AI, 2025)

OSWorld

Complete tasks on real desktop operating systems

WebArena • GAIA

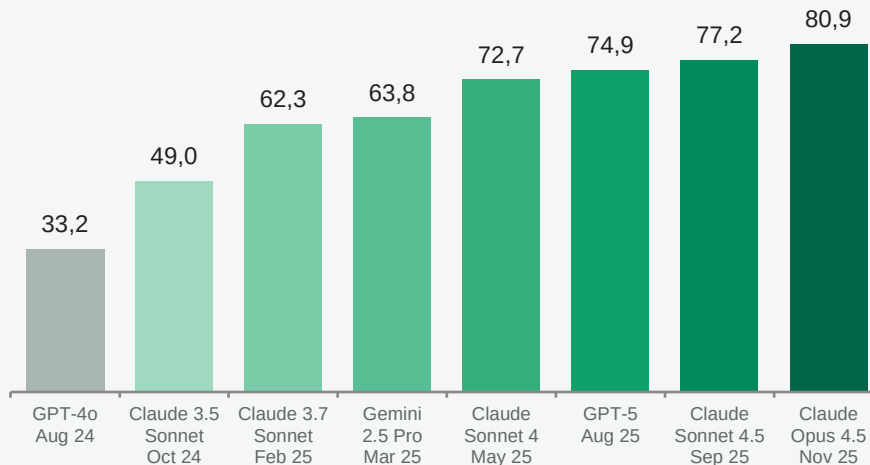
Web tasks • general-assistant questions with tools

τ-bench

Customer-service dialogues with tools and policies

Caveat: in Feb 2026 OpenAI stopped reporting SWE-bench Verified (contamination, flawed tests) and now recommends SWE-bench Pro.

SWE-bench Verified, Aug 2024 – Nov 2025 (% resolved)



Vendor-reported at launch (GPT-5 on 477 of 500 tasks). Scaffolds differ: compare trends, not decimals.

Momentum: capability and adoption

~7 mo

Doubling time of the length of tasks frontier agents complete with 50% reliability (2019–2025)

METR, Mar 2025

<5% → 40%

Enterprise apps integrating task-specific AI agents, 2025 → end of 2026 (forecast)

Gartner, Aug 2025

2% → ~30%

Share of enterprise app software revenue driven by agentic AI, 2025 → 2035 (best case, >\$450B)

Gartner, Aug 2025

GARTNER'S EXPECTED EVOLUTION OF ENTERPRISE AI

2025

AI assistants embedded in almost every enterprise app

2026

Task-specific agents in 40% of enterprise apps

2029

Multi-agent ecosystems collaborating across apps

Where agents are creating value



Software engineering

Navigate codebases, write & test code, fix bugs, open pull requests



Customer service

Resolve requests end-to-end with tools, escalate to humans when needed



Research & analysis

Search many sources, cross-check them and write cited reports



IT & security ops

Triage alerts and incidents, gather evidence, propose remediation



Finance & back-office

Process documents, reconcile records, prepare reports



Sales & marketing

Research accounts, qualify leads, personalise outreach

SECTION

06

Doing it right

How do we deploy agents safely — and create value?



Why reliability is the hard part

Errors compound over long tasks

If every step succeeds independently with probability p , a task of n steps succeeds with probability:

$$P(\text{success}) = p^n$$

- 95% per step → only 36% after 20 steps
- Real steps are not fully independent — this is a simplified model
- Remedy: verification, checkpoints, retries and human review

Key risks and challenges



Prompt injection

Malicious instructions hidden in web pages, documents or emails hijack the agent. #1 in the OWASP Top 10 for LLM Apps (2025).



Excessive agency

Too many permissions or too much autonomy turn a small error into real damage (OWASP LLM06:2025).



Confident errors

Hallucinated facts or wrong actions, amplified over long multi-step tasks.



Cost & latency

Many model calls per task: agents use far more tokens and time than chat.



Accountability

Who approved what? Actions must be traceable, auditable and reversible.



Data exposure

Agents with broad data access can leak sensitive information (OWASP LLM02:2025).

40%+

of agentic AI projects will be canceled by end of 2027 — due to rising costs, unclear business value or weak risk controls.

~130

genuine agentic vendors among thousands — the rest is “agent washing”.

Gartner, June 2025

Design principles for reliable agents



BUILD

● Start simple

A single LLM call or workflow first; add autonomy only when it measurably helps.

● Invest in tool design

Clear names, descriptions, examples and error messages.

● Evaluate continuously

Task-level evals built from real cases, run before every change.



SECURE

● Least privilege

Minimum permissions; sandbox code execution and browsing.

● Treat inputs as untrusted

Separate instructions from data to limit prompt injection.

● Gate risky actions

Human approval for irreversible, costly or sensitive steps.



OPERATE

● Set limits

Cap steps, time and budget; define clear stop conditions.

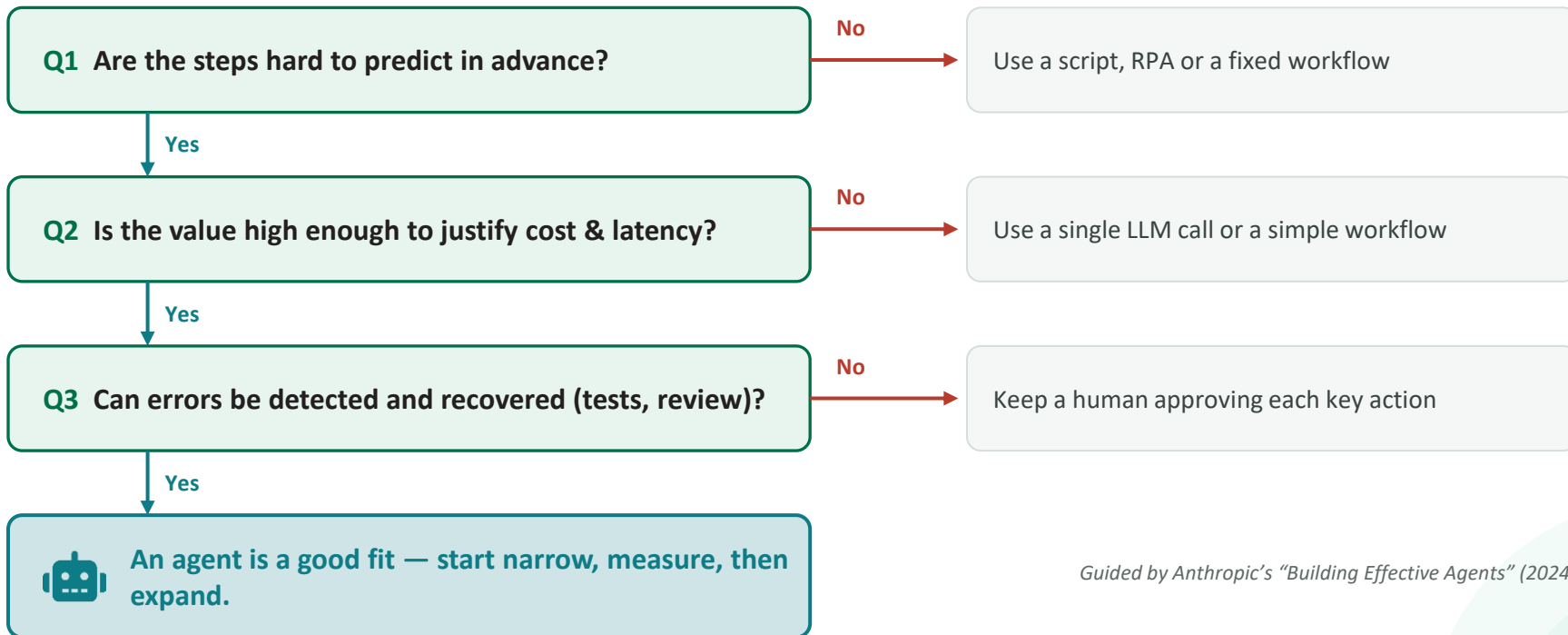
● Make it observable

Trace every reasoning step, tool call and result.

● Plan for failure

Prefer reversible actions, checkpoints and a rollback path.

Do you really need an agent?



Guided by Anthropic's "Building Effective Agents" (2024)

KEY TAKEAWAYS

Agents turn AI from a tool you use into a system that works for you.



An agent is an LLM using tools in a loop, toward a goal.



“Agentic” is a spectrum — choose the least autonomy that works.



Open standards (MCP, A2A) are making agents interoperable.



Value comes with discipline: evals, guardrails, human oversight.

Thank you · Questions?

Sources & further reading

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- Anthropic — “Building Effective Agents” (Dec 2024)
- Anthropic — “How we built our multi-agent research system” (Jun 2025)
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- Yao et al. — ReAct: Synergizing Reasoning and Acting (2022)
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- Yao et al. — Tree of Thoughts (2023)

- Wang et al. — Plan-and-Solve Prompting (2023)
- Sumers et al. — Cognitive Architectures for Language Agents (2023)
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- Jimenez et al. — SWE-bench (2023); OpenAI — SWE-bench Verified (2024) and its retirement note (Feb 2026)
- Gartner press releases — 25 Jun 2025 & 26 Aug 2025
- Linux Foundation — Agentic AI Foundation announcement (9 Dec 2025)
- OWASP — Top 10 for LLM Applications (2025)
- modelcontextprotocol.io · a2a-protocol.org